Motivating example: "8 schools"

$$y_{i,j} \overset{ind}{\sim} \mathcal{N}(\theta_j, \sigma_j^2) \quad i = 1 \ldots n_j, \quad j = 1 \ldots M$$

There are M schools, each with n_j students. We assume that the test scores of the students **within** a school are iid from a shared normal.

Prior over school mean:

$$\theta_j \overset{iid}{\sim} \mathcal{N}(\mu, \tau^2) \quad j = 1 \ldots M$$

We will assume a conjugate prior over the schools' mean test score.

The "**complete conditional**" for the jth school mean is:

$$p(\theta_j \mid -) \equiv p(\theta_j \mid (y_{j,1} \ldots y_{j,n_j})_{j=1}^{M}, (\theta_k)_{k \neq j}, (\sigma_j)_{j=1}^{M}, \tau^2, \mu)$$

"all other variables" — all data — all school means **except** the jth — all other parameters

By conjugacy, this equals:

$$= \mathcal{N}(\theta_j \mid \hat{\theta}_j, V_j)$$

posterior mean $\quad \hat{\theta}_j = \rho_j \bar{y}_j + (1 - \rho_j)\mu$

posterior variance $\quad V_j = \rho_j \frac{\sigma_j^2}{n_j} = \left[\frac{1}{\tau^2} + \frac{n_j}{\sigma_j^2}\right]^{-1}$

$$\text{where } \rho_j = \frac{n_j/\sigma_j^2}{1/\tau^2 + n_j/\sigma_j^2}$$

Notice how the posterior mean changes as a function of the number of observations, the likelihood variance, and the prior variance of the mean.

$n_j \to \infty$ or $\sigma_j^2 \to 0$ or $\tau^2 \to \infty$: $\hat{\theta}_j = \hat{\theta}_j^{MLE}$

$n_j \to 0$ or $\sigma_j^2 \to \infty$ or $\tau^2 \to 0$: $\hat{\theta}_j = \mu$

Notice how the posterior parameters only involve mean (and number) of observations. This is an example of **sufficiency**: the mean and number are **sufficient statistics** for the school mean—i.e.,:

$$P(\theta_j \mid \underbrace{(y_{j,1} \dots y_{j,n_j})_{j=1}^{M}}_{\text{all data}}, -) = p(\theta_j \mid \bar{y}_j, n_j, -)$$

This means the exact same posterior would emerge if we only ever generated and conditioned on the empirical mean. The mean of iid Gaussians is Gaussian:

$$\bar{y}_j \mid \theta_j \overset{\text{ind.}}{\sim} \mathcal{N}(\theta_j, \sigma_j^2/n_j), \quad j = 1 \dots M$$

This is an alternative likelihood leading to the exact same complete conditional. In other words, the two likelihoods are proportionally the same:

$$\prod_{j=1}^{M} \prod_{i=1}^{n_j} \mathcal{N}(y_{i,j} \mid \theta_j^2, \sigma_j^2) \propto_{\theta_j} \prod_{i=1}^{n_j} \exp\left(-\frac{1}{2\sigma^2}(y_{ij} - \theta_j)^2\right)$$

$$\propto_{\theta_j} \prod_{i=1}^{n_j} \exp\left(\frac{1}{\sigma^2} y_{ij}\theta_j - \frac{1}{2\sigma^2}\theta_j^2\right)$$

$$\propto_{\theta_j} \exp\left(\frac{n_j}{\sigma_j^2}\bar{y}_j \theta_j - \frac{n_j}{2\sigma_j^2}\theta_j^2\right)$$

$$\propto_{\theta_j} \mathcal{N}(\bar{y}_j \mid \theta_j, \sigma_j^2/n_j)$$

Integrating out the school-specific means, gives the following marginal likelihood:

$$\int d\theta_1 \dots \int d\theta_M \prod_{j=1}^{M} \mathcal{N}(\bar{y}_j \mid \theta_j, \sigma_j^2/n_j)\, \mathcal{N}(\theta_j \mid \mu, \tau^2)$$

$$= \prod_{j=1}^{M} \mathcal{N}(\bar{y}_j \mid \mu, \tau^2 + \sigma^2/n_j)$$

This is a "normal means" model, with each mean having one observation.

$$\bar{y}_j \overset{\text{ind.}}{\sim} \mathcal{N}(\mu, \tau^2 + \sigma^2/n_j), \quad j = 1 \dots M$$

Empirical Bayes, type-II MLE, and shrinkage estimators

Using the above model, one could imagine **fitting** the prior mean via **MLE**:

$$\hat{\mu}^{MLE} = \sum_{j=1}^{M} \left[\frac{w_j}{\sum_k w_k} \right] \bar{Y}_j \,, \qquad w_j = \frac{1}{\tau^2 + \sigma_j^2/n_j}$$

In this case, this is called **type-II MLE**, or maximum **marginal** likelihood estimation since we are maximizing with respect to the marignal likelihood (with the means integrated out).

Consider again the posterior mean of the school-specific means, but using the fitted prior mean:

$$\hat{\theta}_j = p_j \bar{Y}_j + (1-p_j)\hat{\mu}^{MLE}$$

This is a **shrinkage estimator** since we are shrinking the different estimates towards the fitted mu. Such estimators can have lower risk than estimating each mean according to the local **MLE**s, even if estimates are shrunk towards an arbitrary location (e.g., see the James-Stein paradox).

This is also an example of an **empirical Bayes** procedure where we set the priors in a data-driven way. There is a close connection between shrinkage estimators and empirical Bayes.

The prior variance plays an important role of controlling how observations are **pooled**:

As the variance gets very large, the weights are (nearly) equal across all schools (this isn't rigorous, the limit is technically undefined), and we get a **completely pooled** estimate. This corresponds to the prior assumption that school means do not bear any information on each other.

$$\tau^2 \to \infty : \quad \frac{w_j}{\sum_k w_k} \to \frac{1}{M} \quad \text{thus} \quad \hat{\mu}^{MLE} \to \frac{1}{M}\sum_{j=1}^{M} \bar{Y}_j \qquad \textbf{no pooling}$$

As the prior variance goes to 0, the prior Gaussian goes to a delta spike:

$$\tau^2 \to 0 : \quad \theta_j \sim \delta_\mu \qquad \text{i.e.,} \quad \theta_j = \mu \quad \text{almost surely}$$

In other words, it is consistent with the assumption that all schools have the same mean (equal to mu).

$$\tau^2 \to 0 : \quad \hat{\mu}^{MLE} = \sum_{j=1}^{M} \frac{n_j/\sigma_j^2}{\sum_k n_k/\sigma_k^2} \bar{Y}_j = \sum_{j=1}^{M} \frac{1}{\sum_k n_k/\sigma_k^2} \sum_{i=1}^{n_j} Y_{i,j}$$

$$\text{Say } \sigma_j^2 = \sigma^2 \text{ is constant :} \quad = \frac{1}{n}\sum_j \sum_{i=1}^{n_j} Y_{i,j} \qquad \textbf{complete pooling}$$

Hierarchical modeling

The prior mean μ plays the role of **shrinkage location** for the means across schools. The prior variance τ^2 then determines the **degree of pooling.**

One data-adaptive way to set these would be to do type-II MLE (i.e., empirical Bayes). Another is to place priors and infer them as latent variables (i.e., hierarchical modeling)

$$\hat{\theta}_j = \mathbb{E}[\theta_j \mid y_j, \hat{\mu}_{MLE}, \tau^2_{MLE}]$$ posterior mean w/ empirical Bayes

vs.

$$\hat{\theta}_j = \mathbb{E}_{(\mu,\tau^2)\sim p(\mu,\tau^2 \mid y)}\left[\mathbb{E}[\theta_j \mid y_j, \mu, \tau^2]\right]$$ posterior mean w/ hierarchical model

posterior of hyperparameters

A conjugate prior would let us get the posterior of hyperparameters in closed form.
A conjugate prior in this case would have the following form:

$$(\mu, \tau^2) \sim g(\eta_0)$$

$$\theta_j \overset{iid}{\sim} \mathcal{N}(\mu, \tau^2)$$

$$p(\mu, \tau^2 \mid \theta_1 \dots \theta_M) = g(\mu, \tau^2 \mid \eta_M)$$

The conjugate prior is the **normal-inverse (scaled) chi-squared (NIX) distribution:**

$$g(\mu, \tau^2 \mid \eta_0) = \mathrm{NIX}(\mu, \tau^2 \mid \nu_0, \tau_0^2, \mu_0, \kappa_0)$$
$$= \mathcal{N}(\mu \mid \mu_0, \tau^2/\kappa_0)\, \chi^{-2}(\tau^2 \mid \nu_0, \tau_0^2)$$

This is a **bivariate** distribution equal to an inverse scaled chi-squared times a **conditional** Gaussian.

Due to conjugacy, the **joint** posterior is another NIX distribution:

$$p(\mu, \tau^2 \mid \theta_1 \dots \theta_M, \eta_0) = g(\mu, \tau^2 \mid \eta_M) = \mathrm{NIX}(\mu, \tau^2 \mid \nu_M, \tau_M^2, \mu_M, \kappa_M)$$

for certain **posterior parameters** whose form is left unstated. (You will derive this in HW1.)

Conjugate prior(s) for variance of Gaussian likelihoods with known mean

The conjugate prior for the mean of a Gaussian likelihood with **known** variance is Gaussian.
How about for the reverse? The variance must be positive, so the conjugate prior must be some continuous distribution with support >0. The answer is the **inverse scaled chi-squared distribution.**

$$y_i \sim \mathcal{N}(\mu, \tau^2)$$ likelihood (with known mean)

$$\tau^2 \sim \chi^{-2}(\nu_0, \tau_0^2)$$ conjugate prior for variance

"degrees of freedom" (ν_0)

"scale" (sometimes called "mean" loosely) (τ_0^2)

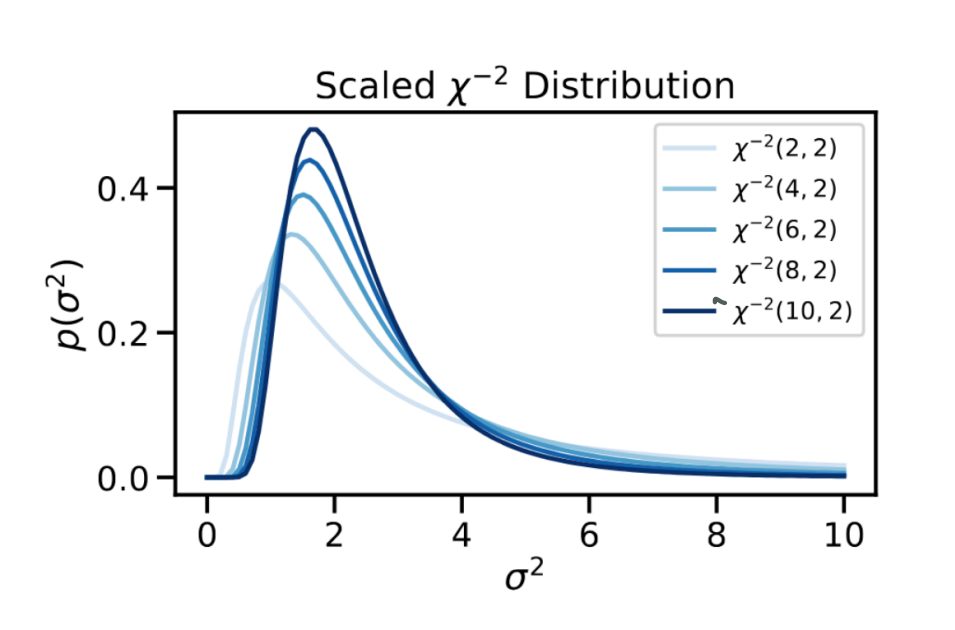
The inverse chi-squared is asymmetric with mode < mean, and scale between them.

$$\frac{\nu_0}{\nu_0+2}\tau_0^2 \leq \tau_0^2 \leq \frac{\nu_0}{\nu_0-2}\tau_0^2$$

mode

mean $\mathbb{E}[\tau^2]$

$$\chi^{-2}(\tau^2 \mid \nu_0, \tau_0^2) = \frac{\left(\frac{\nu_0 \tau_0^2}{2}\right)}{\Gamma\left(\frac{\nu_0}{2}\right)} [\tau^2]^{-\frac{\nu_0}{2}-1} \exp\left(-\frac{1}{2\tau^2}\nu_0\tau_0^2\right)$$



It is a reparameterization of the **inverse gamma distribution:**

"shape" "rate"

$$\chi^{-2}(\nu_0, \tau_0^2) \equiv \text{InvGamma}\left(\frac{\nu_0}{2}, \frac{\nu_0\tau_0^2}{2}\right)$$

The inverse of an inverse chi-squared random variable is a chi-square random variable.
This is a conjugate prior for the **precision** of a Gaussian likelihood with known mean.

$$\lambda \overset{\text{def}}{=} 1/\tau^2 \qquad \longrightarrow \qquad \lambda \sim \chi^2(\nu_0, \lambda_0)$$

$$\lambda_0 \overset{\text{def}}{=} 1/\tau_0^2$$

"degrees of freedom" "scale"

The (scaled) chi-squared distribution is a reparameterization of the **gamma distribution.**

$$\chi^2(\lambda; \nu_0, \lambda_0) \equiv \mathrm{Ga}\left(\lambda; \frac{\nu_0}{2}, \frac{\nu_0}{2\lambda_0}\right)$$

"shape" "rate"

Inverse chi-squared—Gaussian conjugacy:

$$\tau^2 \sim \chi^{-2}(\nu_0, \tau_0^2)$$

$$\theta_j \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \tau^2) \qquad j = 1 \ldots M$$

$$p(\tau^2 \mid -) \propto_{\tau^2} \chi^{-2}(\tau^2 \mid \nu_0, \tau_0^2) \prod_{j=1}^{M} \mathcal{N}(\theta_j \mid \mu, \tau^2)$$

$$\propto [\tau^2]^{-\frac{\nu_0}{2}-1} \exp\left(-\frac{1}{2\tau^2}\nu_0\tau_0^2\right) \prod_{j=1}^{M} \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{1}{2\tau^2}(\theta_j-\mu)^2\right)$$

$$\propto [\tau^2]^{-\frac{\nu_0}{2}-\frac{M}{2}-1} \exp\left(-\frac{1}{2\tau^2}\underbrace{\left[\nu_0\tau_0^2 + \sum_{j=1}^{M}(\theta_j-\mu)^2\right]}_{\equiv \nu_M \tau_M^2}\right)$$

$$\propto \chi^{-2}(\tau^2 \mid \nu_M, \tau_M^2)$$

$$\nu_M = \nu_0 + M$$

$$\tau_M^2 = [\nu_0 + M]^{-1}\left[\nu_0\tau_0^2 + \sum_{j=1}^{M}(\theta_j-\mu)^2\right]$$

$$\tau_m^2 = \left(\frac{\nu_0}{\nu_0+m}\right)\tau_0^2 + \left(\frac{m}{\nu_0+m}\right)\underbrace{\frac{1}{m}\sum_{j=1}^{m}(\theta_j-\mu)^2}_{=\hat{\tau}^2_{MLE}}$$

Notice that only the sum-of-squared errors and M enter the posterior parameters. This is another instance of **sufficiency**. WLOG say mu=0, then we have:

$$p(\tau^2 \mid \theta_1 \ldots \theta_m, -) = p(\tau^2 \mid m, \bar{\theta}_j^2)$$

This means if we only generated and conditioned on the sum-of-squares, the posterior would be the same. The distribution of this statistic is chi-squared:

$$\theta_j \overset{iid}{\sim} \mathcal{N}(0, \tau^2), \quad j = 1 \ldots m \quad \Big\| \rightarrow \quad \bar{\theta}^2 \sim \chi^2(m, \tau^2)$$

$$\bar{\theta}_j^2 = \sum_j \theta_j^2$$

Let's confirm we would get the same posterior:

$$p(\tau^2 \mid \bar{\theta}^2, m, -) \propto_{\tau^2} \chi^{-2}(\tau^2 \mid \nu_0, \tau_0^2)\, \chi^2(\bar{\theta}^2 \mid m, \tau^2)$$

$$\propto [\tau^2]^{-\frac{\nu_0}{2}-1} \exp\left(-\frac{1}{2\tau^2}\nu_0\tau_0^2\right) \left[\frac{m}{2\tau^2}\right]^{m/2} \exp\left(-\bar{\theta}^2 \frac{m}{2\tau^2}\right)$$

$$\propto [\tau^2]^{-\frac{\nu_0}{2}-\frac{m}{2}-1} \exp\left(-\frac{1}{2\tau^2}\left[\nu_0\tau_0^2 + \sum_{j=1}^{m}\theta_j\right]\right) \quad \checkmark$$

This not only confirms that this is a sufficient statistic, but also shows that the inverse chi-squared is also a conjugate prior to the chi-squared distribution!

Intro to probabilistic graphical models (PGMs):

Consider the following hierarchical model for the 8 schools data:

$$\tau^2 \sim \mathcal{N}^{-2}(\nu_0, \tau_0^2)$$

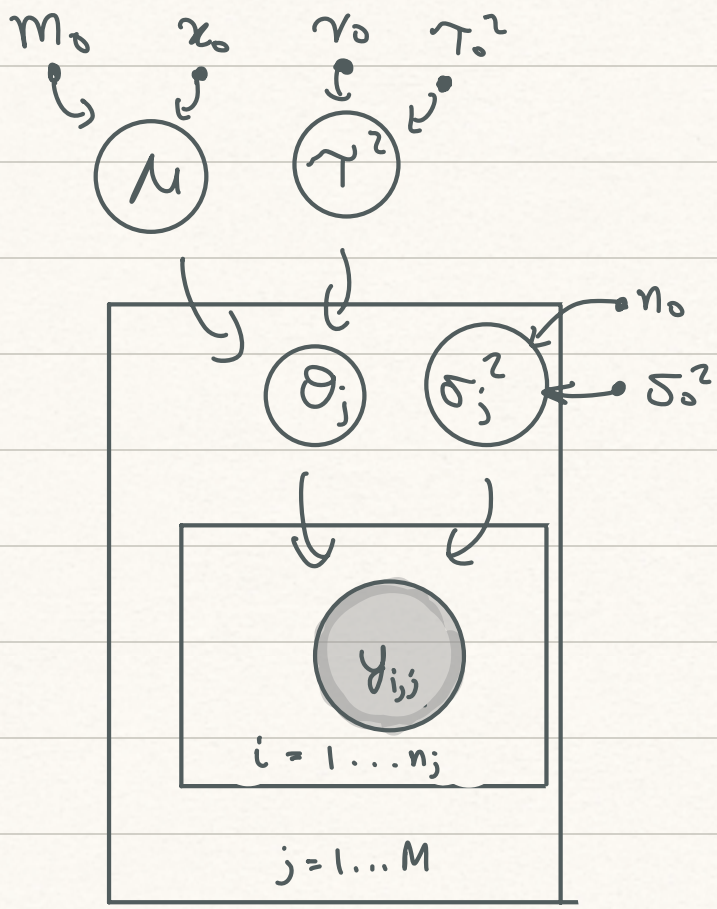
$$\mu \sim \mathcal{N}(m_0, 1/\kappa_0)$$

$$\sigma_j^2 \overset{iid}{\sim} \chi^{-2}(n_0, \sigma_0^2) \qquad j = 1 \ldots M$$

$$\theta_j \overset{iid}{\sim} \mathcal{N}(\mu, \tau^2) \qquad j = 1 \ldots M$$

$$y_{i,j} \overset{ind}{\sim} \mathcal{N}(\theta_j, \sigma_j^2) \qquad j = 1 \ldots M, \quad i = 1 \ldots n_j$$

This is actually still a simple model, but it's already hard to read. PGMs help us express and visualize hierarchical models.



This is a **probabilistic graphical model (PGM)** corresponding to the generative process above.

The **shaded nodes** are **observed** variables, while the **unshaded nodes** are **latent** variables.

The filled nodes are **hyperparameters** which are not random (i.e., fixed and usually known).

The **plates** denote **repeated sampling**.

$$Z = \{\mu, \tau^2, \theta_1 \dots \theta_M, \sigma^2_{1 \dots M}\}$$

the set of all latent variables in the model

The joint distribution over all latent and observed variables can always be written as:

$$P(Z, Y) = \left[\prod_{d=1}^{D} P(Z_d \mid Z_{<d})\right] P(Y \mid Z)$$

However, this does not take into account any conditional independences in the model.

The joint distribution described by a PGM can always be factorized as:

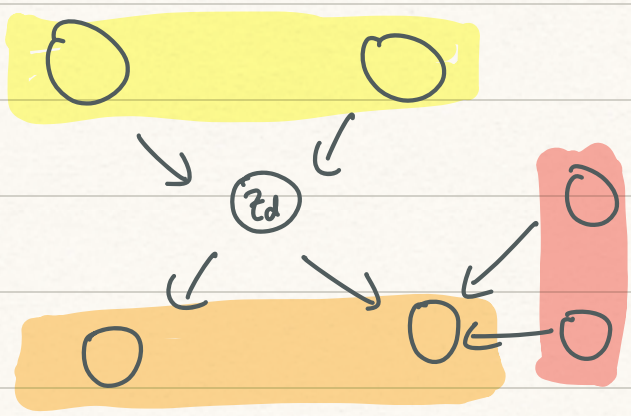
$$= \left[\prod_{d=1}^{D} P(Z_d \mid Z_{par(Z_d)})\right] P(Y \mid Z_{par(Y)})$$

where **par(Zd)** is the set of all nodes in the graph that are parents of node Zd.

More generally, PGMs describe the set of all conditional independences in a joint distribution. A node is conditionally independent of all other nodes given its **Markov blanket.**

$$MB(Z_d) = \{Par(Z_d) \cup Child(Z_d) \cup Co\text{-}par(Z_d)\}$$

"co-parents"



$$Z_d \perp\!\!\!\perp Z_{d'} \mid MB(Z_d) \quad \forall\, Z_{d'} \notin MB(Z_d)$$

Preview to Gibbs sampling and MCMC:

Typically we interact with the posterior via **posterior expectations.**

For example, above we motivated the posterior expectation of the school means:

$$\hat{\theta}_j = \mathbb{E}_{p(\theta_j \mid y)}[\theta_j]$$

where the expectation is with respect to the **posterior marginal:**

$$p(\theta_j \mid y) = \int p(\theta_j, z_{\setminus \theta_j} \mid y)\, dz_{\setminus \theta_j}$$

posterior marginal

marginalizes out all **other** latent

In complicated enough models, we cannot analytically compute these marginals.

However, we can still **approximate posterior expectations** using Monte Carlo.

$$\mathbb{E}_{p(\theta_j \mid y)}[f(\theta_j)] = \frac{1}{S}\sum_{s=1}^{S} f(\theta_j^s)$$

$$\theta_j^s \overset{iid}{\sim} p(\theta_j \mid y), \quad s = 1 \dots S$$

If we cannot analytically compute the marginal posterior, how do we sample from it?

Gibbs sampling:

for $s = 1 \dots S$:

for $d = 1 \dots D$:

$z_d \sim p(z_d \mid z_{\setminus d}, y)$ re-sample from complete conditional

$z_d^s = z_d$ save sample

<u>claim</u>: $\lim_{S \to \infty} \Pr(z_d^s) = p(z_d \mid y)$

The takeaway: even if the joint posterior over all latent variables is not tractable, or the desired posterior marginal is not tractable, we can approximate the marginal using Gibbs sampling as long as the **complete conditional are tractable.**

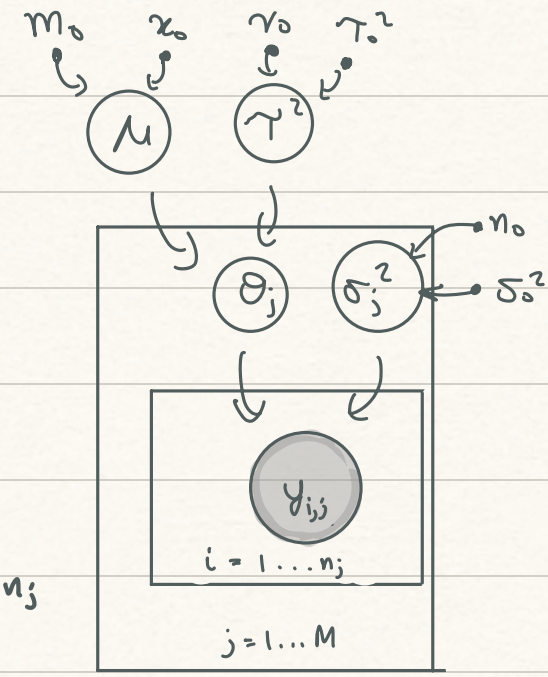
$$\tau^2 \sim \mathcal{N}^{-2}(\nu_0, \tau_0^2)$$

$$\mu \sim \mathcal{N}(m_0, 1/\kappa_0)$$

$$\sigma_j^2 \overset{iid}{\sim} \chi^{-2}(n_0, \sigma_0^2) \qquad j = 1 \ldots M$$

$$\theta_j \overset{iid}{\sim} \mathcal{N}(\mu, \tau^2) \qquad j = 1 \ldots M$$

$$y_{i,j} \overset{ind}{\sim} \mathcal{N}(\theta_j, \sigma_j^2) \qquad j = 1 \ldots M, \quad i = 1 \ldots n_j$$



For example, the complete conditional for the likelihood variance follows from conjugacy, since conditioning on all other variables turns the problem into an inverse chi-squared prior over the variance of a Gaussian likelihood with **known** mean.

e.g., $$p(\sigma_j^2 \mid -) = \chi^{-2}(\sigma_j^2 \mid \alpha_{n_j}, \sigma_{n_j}^2)$$

$$\alpha_{n_j} = \alpha_0 + n_j, \quad \sigma_{n_j}^2 = [\alpha_0 + n_j]^{-1} \left[\alpha_0 \sigma_0^2 + \sum_{i=1}^{n_j} (y_{ij} - \theta_j)^2\right]$$

As another example, the complete conditional for mu follows from Gaussian-Gaussian conjugacy, where here the school means are treated like data:

e.g. $$p(\mu \mid -) = \mathcal{N}(\mu \mid m_M, 1/\kappa_M)$$

$$m_M = \rho_M \bar{\theta} + (1 - \rho_M) m_0, \quad \kappa_M = \kappa_0 + 1/\tau^2$$

$$\rho_M = \frac{1/\tau^2}{\kappa_0 + 1/\tau^2}$$

Notice that the complete conditional for mu involves tau^2, but not sigma^2. Why is that? This has to do with conditional independences in the model. PGMs make these easy to see.